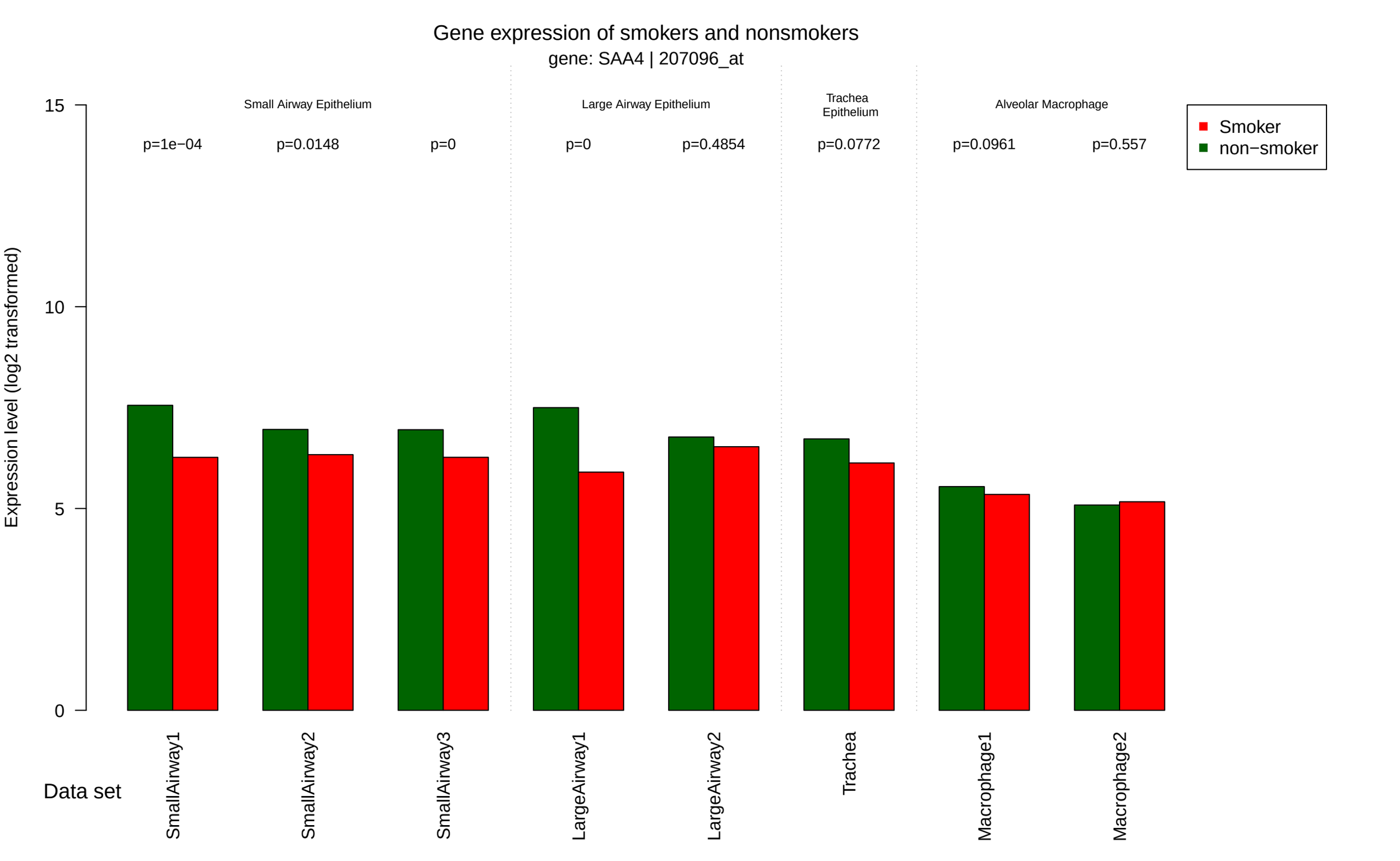
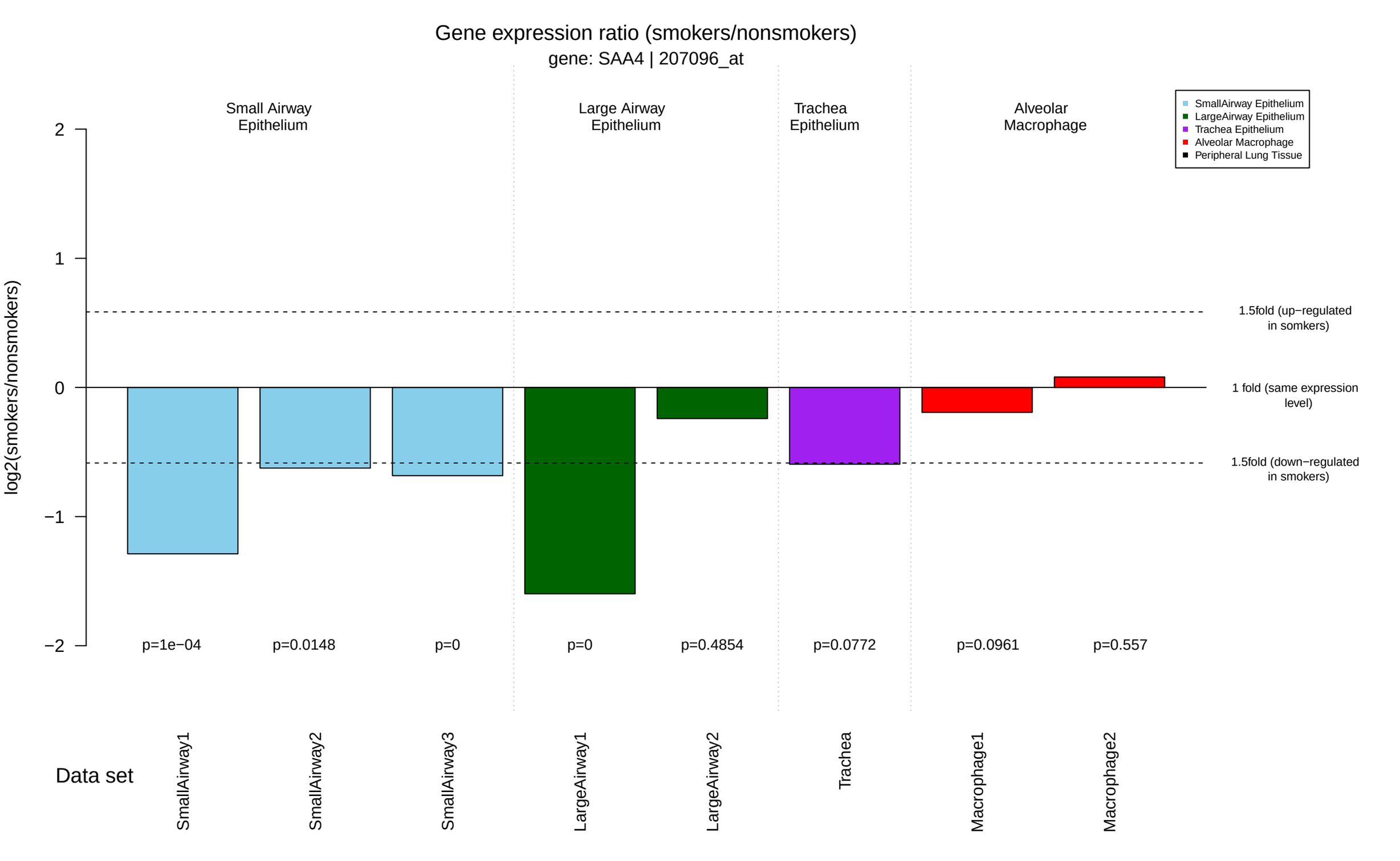
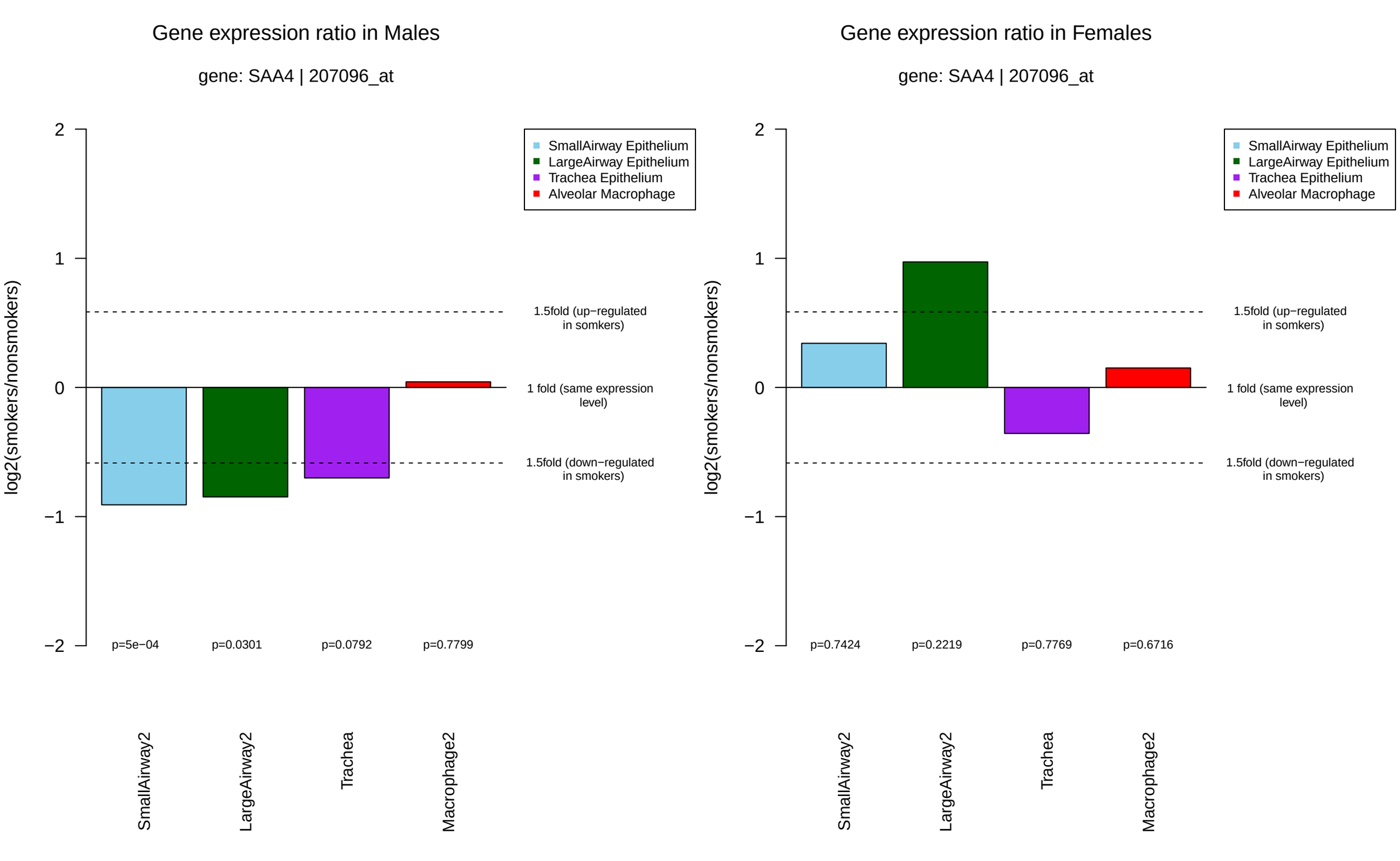
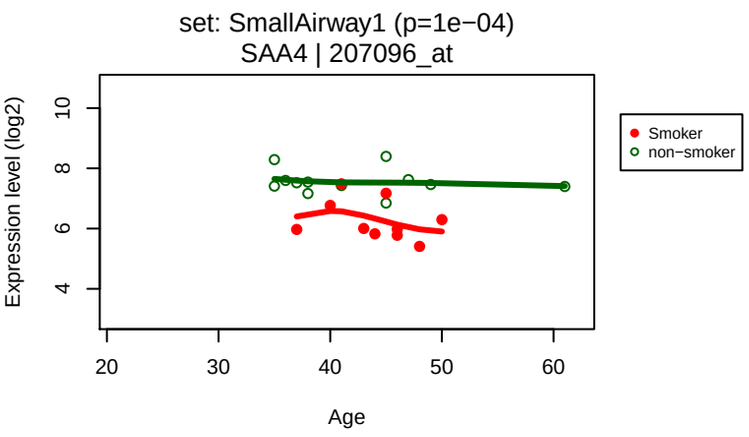
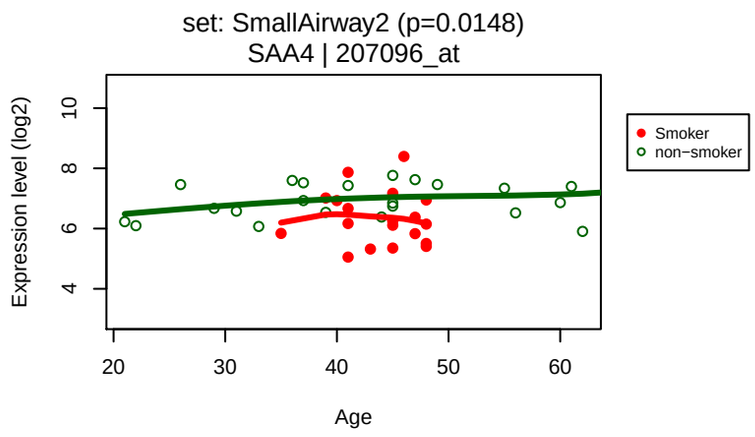
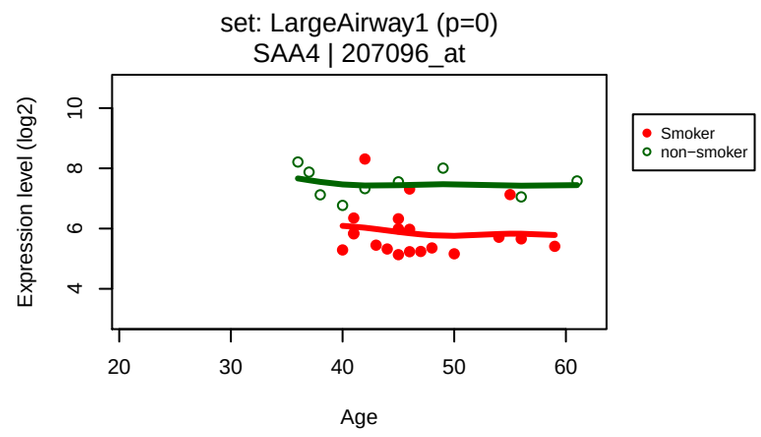

Gene expression of smokers and nonsmokers
gene: SAA4 | 207096_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=1e−04
p=0.0148
p=0
p=0
p=0.4854
p=0.0772
p=0.0961
p=0.557
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

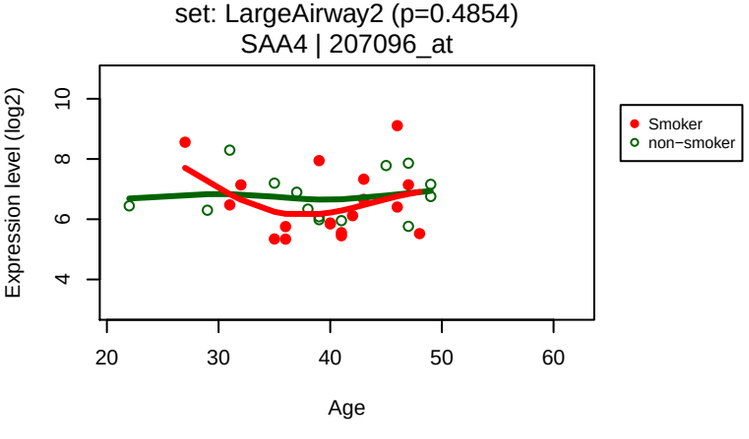

Gene expression ratio (smokers/nonsmokers)
gene: SAA4 | 207096_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0148
p=0
p=0
p=0.4854
p=0.0772
p=0.0961
p=0.557
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

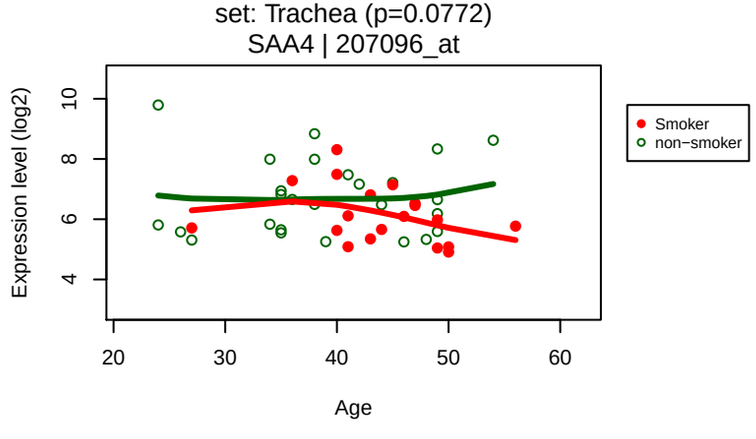

Gene expression ratio in Males
gene: SAA4 | 207096_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.0301
p=0.0792
p=0.7799
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAA4 | 207096_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7424
p=0.2219
p=0.7769
p=0.6716
SmallAirway2
LargeAirway2
Trachea
Macrophage2

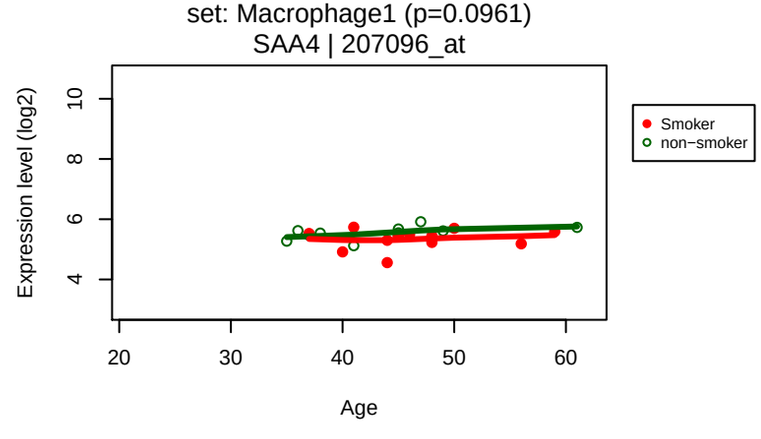

set: SmallAirway1 (p=1e−04)
SAA4 | 207096_at
Expression level (log2)
Age
Smoker
non−smoker

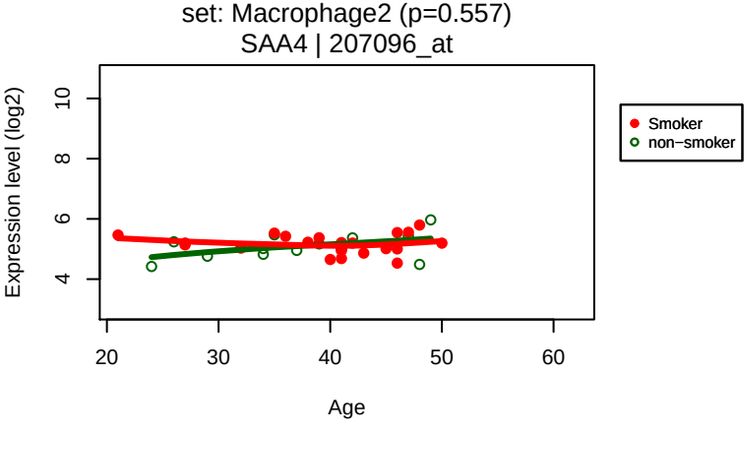

set: SmallAirway2 (p=0.0148)
SAA4 | 207096_at
Expression level (log2)
Age
Smoker
non−smoker

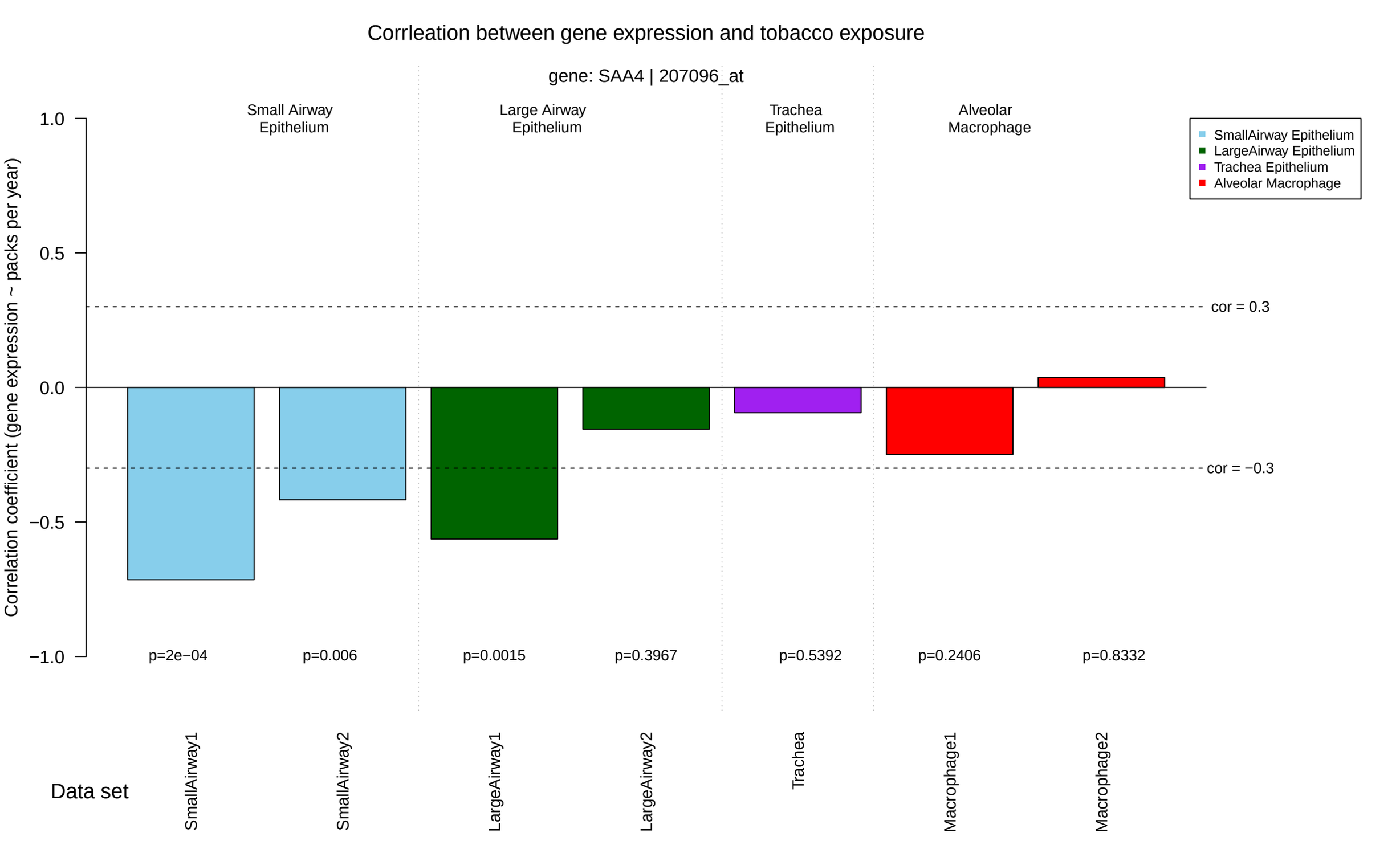

set: LargeAirway1 (p=0)
SAA4 | 207096_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4854)
SAA4 | 207096_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0772)
SAA4 | 207096_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0961)
SAA4 | 207096_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.557)
SAA4 | 207096_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAA4 | 207096_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=2e−04
p=0.006
p=0.0015
p=0.3967
p=0.5392
p=0.2406
p=0.8332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2